Interpolare

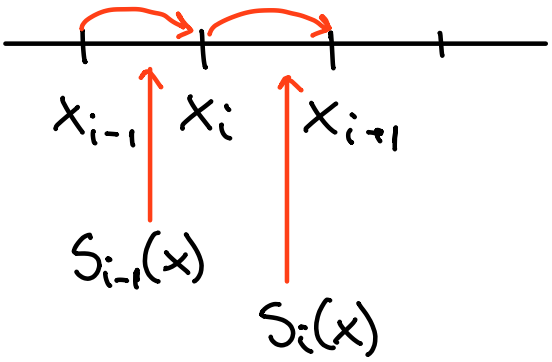
x_i ; y_i $\quad f(x) = ?$ $\quad P_n(x) = \sum_{i=0}^{n} a_i u_i(x) = \sum_{i=0}^{n} y_i P_i(x)$

— Lagrange l_i

— Newton $\mathcal{F}$

$P_n(x_i) = y_i \longrightarrow$ condiție de interpolare

Interpolare cu funcții spline



$$S(x) = \begin{cases} S_0(x) & x \in [x_0, x_1) \\ S_1(x) & x \in [x_1, x_2) \\ S_2(x) & x \in [x_2, x_3) \\ \vdots & \\ S_{n-1}(x) & x \in [x_{n-1}, x_n) \end{cases}$$

Se dau puncte $(x_0\ y_0), (x_1\ y_1) \ldots (x_n\ y_n)$

$z_0 \quad z_1 \quad z_2 \ldots z_4$

$P_{01} \quad P_{02}$

Boundry conditions

$x_{i-1} \quad x_i \quad x_{i+1}$

$P \quad Q$

$P(x_i) = Q(x_i)$

$P'(x_i) = Q'(x_i)$

$P''(x_i) = Q''(x_i) \rightarrow$ optional

ex $(1;2)$ $(2;3)$ $(3;4)$

$ax+b$ $cx+d$

$P(x)=Q_1(x)$ $a+b=2$

$2a+b=2c+d=3 \Rightarrow b=d=1$

$3c+d=4$ $a=c=1$

$P'(x)=Q_1'(x)$ $a=c$

ex $(1;2)$ $(2;3)$ $(3;4)$

ax^2+bx+c dx^2+ex+f

$a+b+c=2$

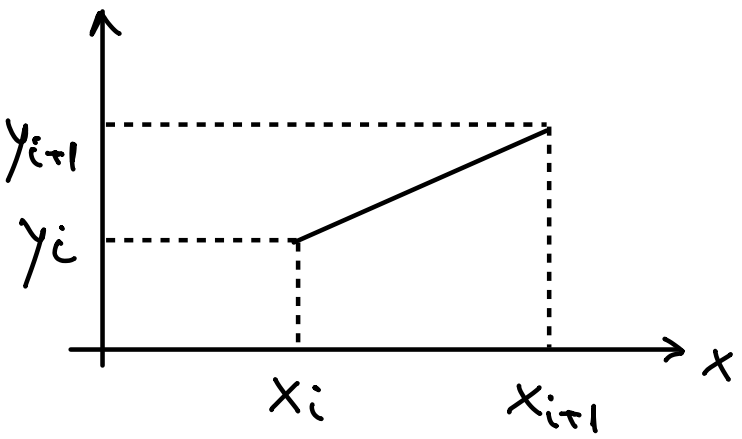
$4a+2b+c=4d+2e+f=3$

$9d+3e+f=4$

$4a+b=4d+e$

fixez derivata intr-un capăt $2a+b=2$

a) Interpolare spline liniară



$S_i(x)=a_i x_i+b_i$

Condiții interpolare $\begin{cases} S_i(x_i) = y_i \\ S_i(x_{i+1}) = y_{i+1} \end{cases} \Rightarrow \begin{cases} a_i x_i + b_i = y_i \\ a_i x_{i+1} + b_i = y_{i+1} \end{cases}$

$a_i(x_{i+1} - x_i) = y_{i+1} - y_i$

$$a_i = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}$$

$$b_i = y_i - a_i x_i$$

b) Interpolare spline cu funcții cubice

b1) de clasă C^1

x	x_0	x_1	x_n
f	y_0	y_1 ...	y_n
f'	f'_0	f'_1	f'_n

$f'_i \overset{not}{=} f'(x_i)$

$$S_i(x) = a_i(x - x_i)^3 + b_i(x - x_i)^2 + c_i(x - x_i) + d_i$$

4 necunoscute

Condiții interpolare $\begin{cases} S_i(x_i) = y_i \\ S_i(x_{i+1}) = y_{i+1} \end{cases}$ 2 ec.

$\begin{cases} S_i'(x_i) = f'_i \\ S_i'(x_{i+1}) = f'_{i+1} \end{cases}$ 2 ec.

b2) de clasă C^2

x	x_0	x_1	....	x_n
f	y_0	y_1	...	y_n

$$S_i(x_i) = y_i$$
$$S_i(x_{i+1}) = y_{i+1}$$
} $2n$

$$S_i'(x_{i+1}) = S_{i+1}'(x_{i+1}) \quad n-1$$
$$S_i''(x_{i+1}) = S_{i+1}''(x_{i+1}) \quad n-1$$

$4n-2$ nec.

funcții spline tensionate
$$\begin{cases} S_0'(x_0) = f_0' \\ S_{n-1}'(x_n) = f_n' \end{cases} \quad 2$$

SAU $\Rightarrow 4n$

funcții spline naturale
$$\begin{cases} S_0''(x_0) = 0 \\ S_{n-1}''(x_n) = 0 \end{cases} \quad 2$$

Problemă examen

x	0 (S_0)	1 (S_1)	2
y	4	-7	8
f'	0	—	2

C^2 tensionate

$$S_0(x) = a_0x^3 + b_0x^2 + c_0x + d_0$$
$$S_1(x) = a_1(x-1)^3 + b_1(x-1)^2 + c_1(x-1) + d_1$$

$S_0(0) = 4$

$S_0(1) = -7$

$S_1(1) = -7$

$S_1(2) = 8$

$S_0'(1) = S_1'(1)$

$S_0''(1) = S_1''(1)$

$S_0'(0) = 0$

$S_1'(2) = 2$

$\boxed{d_0 = 4}$

$a_0 + b_0 + c_0 + d_0 = -8$

$\boxed{d_1 = -7}$

$a_1 + b_1 + c_1 + d_1 = 8$

$3a_0 + 2b_0 + c_0 = c_1$

$6a_0 + 2b_0 = 2b_1$

$\boxed{c_0 = 0}$

$3a_1 + 2b_1 + c_1 = 2$

$a_0 + b_0 = -11$

$a_1 + b_1 + c_1 = 15$

$3a_0 + 2b_0 = c_1$

$3a_0 + b_0 = b_1$

$3a_1 + 2b_1 + c_1 = 2$

$a_1 + 6a_0 + 3b_0 = 15$

$3a_1 + 9a_0 + 4b_0 = 2$

$a_0 + b_0 = -11$

$$\begin{cases} 9a_0 + 5b_0 = 43 \\ 5a_0 + 5b_0 = -55 \end{cases}$$

$\Rightarrow 4a_0 = 98 \Rightarrow \boxed{a_0 = \frac{49}{2}}$

$b_0 = -11 - a_0 \Rightarrow \boxed{b_0 = -\frac{71}{2}}$

$a_1 = 15 - 6a_0 - 3b_0$

$= 15 - \frac{6 \cdot 49 - 3 \cdot (-71)}{2} \Rightarrow \boxed{a_1 = -\frac{51}{2}}$

$\boxed{b_1 = 38}$

$\boxed{c_1 = \frac{5}{2}}$